Name: ____________________________ (please print)

Signature: _________________________

ECE 2202 – Quiz 4

March 25, 2026

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.

2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.

3. You will have 40 minutes to work on the quiz.

4. Write your ID number on every page that you show your work.

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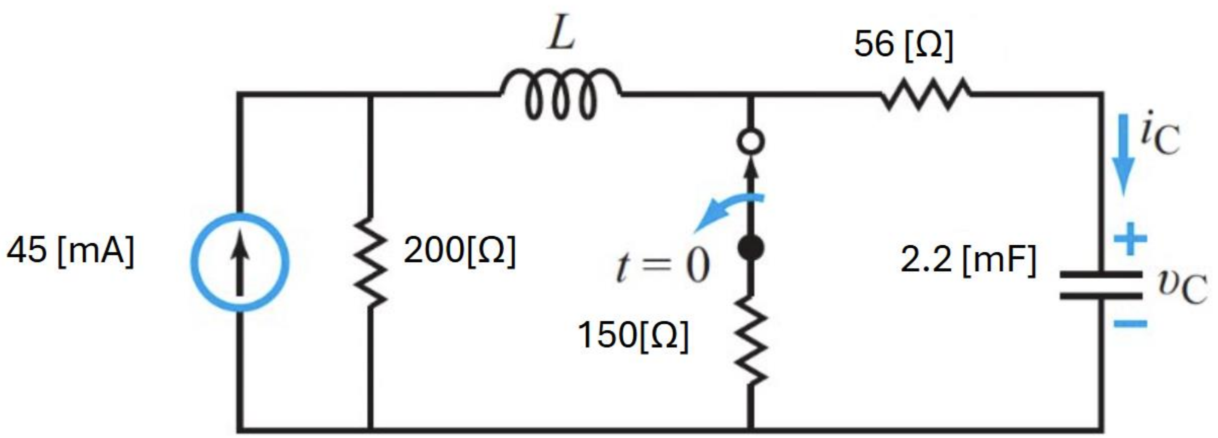
Problem 1/1 (20 pts)

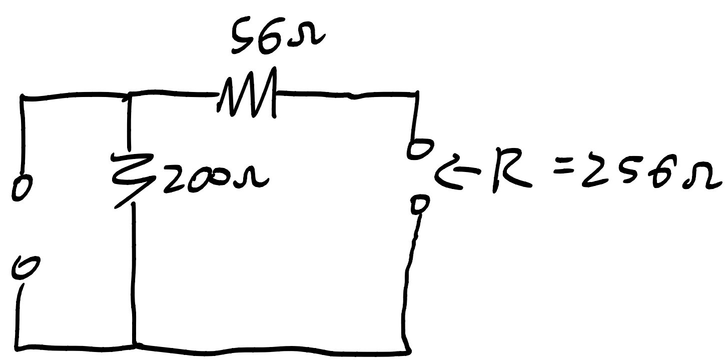
Choose the value of the inductor in the circuit below so that v_c exhibits a critically damped response and determine $v_c(t)$ for $t \geq 0$.



$W_o = \alpha$

$$\frac{1}{\sqrt{LC}} = \frac{R}{2L}$$

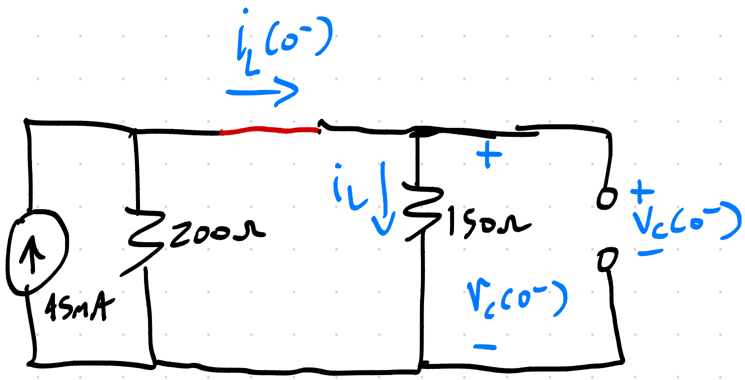
$$\frac{1}{LC} = \frac{R^2}{4L^2}$$



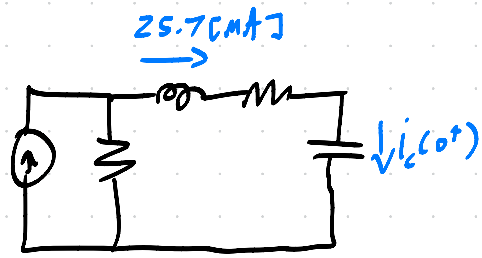
$$\frac{4L}{C} = R^2 \Rightarrow L = \frac{R^2 C}{4} = 36.04\ [H]$$

$$\alpha = \frac{R}{2L} = 3.55\ [s^{-1}]$$

$t=0^-$



$t=0^+$



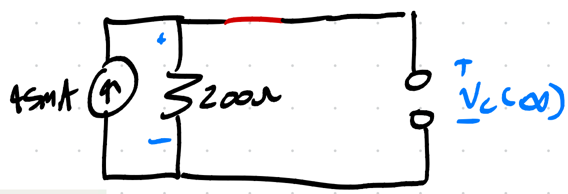
$i_c(0^+) = 25.7\,[mA]$

$$i_L(0^-) = 45\,[mA]\left(\frac{200\Omega}{(150+200)\Omega}\right)$$

$$= 25.7\,[mA]$$

$$V_c(0^-) = 25.7(150) = 3.86\,[V]$$

$t=\infty$



$V_c(\infty) = 45\,[mA]\cdot 200\Omega$

$= 9\,[V]$

Critically Damped $\alpha = \omega_0$

$$x(t) = [(B_1 + B_2 t)e^{-\alpha t} + x(\infty)]\,u(t)$$

$B_1 = x(0) - x(\infty)$ $\qquad$ $B_2 = x'(0) + \alpha[x(0) - x(\infty)]$

$$B_1 = V_c(0) - V_c(\infty) = (3.86 - 9)\,[V] = -5.14\,[V]$$

$$B_2 = V_c'(0) + \alpha(V_c(0) - V_c(\infty))$$

$$= 11.68 + 3.55(-5.14) = -6.57\left[\tfrac{V}{s}\right]$$

$i_c(0) = C\,V_c'(0)$

$\Rightarrow V_c'(0^-) = \dfrac{i_c(0)}{C}$

$= \dfrac{25.7\,[mA]}{2.2\,[mF]} = 11.68\left[\tfrac{V}{s}\right]$

$$V_c(t) = \left(-5.14\,[V] - 6.57\left[\tfrac{V}{s}\right]t\right)e^{-3.55[s^{-1}]t} + 9\,[V] \quad t \geq 0$$

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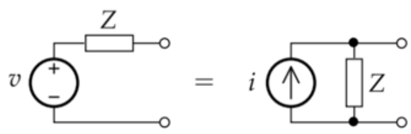
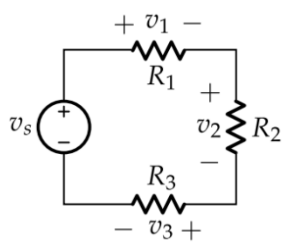
$$i = \frac{\mathrm{d}q}{\mathrm{d}t} \qquad v = \frac{\mathrm{d}w}{\mathrm{d}q} \qquad p = \frac{\mathrm{d}w}{\mathrm{d}t} = \frac{\mathrm{d}w}{\mathrm{d}q}\frac{\mathrm{d}q}{\mathrm{d}t} = iv \qquad R_1||R_2 = \frac{1}{\frac{1}{R_1}+\frac{1}{R_2}} = \frac{R_1R_2}{R_1+R_2}$$



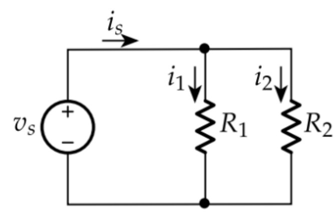
$$v = i \cdot Z$$
$$i = \frac{v}{Z}$$

passive sign convention:
$v = iR$
$p_{abs} = vi$
$p_{del} = -vi$
active sign convention:
$v = -iR$
$p_{del} = vi$
$p_{abs} = -vi$

$$p = i^2R = \frac{v^2}{R}$$



$$v_1 = \frac{R_1}{R_1+R_2+R_3}v_s$$
$$v_2 = \frac{R_2}{R_1+R_2+R_3}v_s$$
$$v_3 = \frac{R_3}{R_1+R_2+R_3}v_s$$



$$i_1 = \frac{R_2}{R_1+R_2}i_s$$
$$i_2 = \frac{R_1}{R_1+R_2}i_s$$

Steps to find Thevenin/Norton Equivalent

$$i = C\frac{\mathrm{d}v}{\mathrm{d}t} \qquad v = L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$p = Cv\frac{\mathrm{d}v}{\mathrm{d}t} \qquad p = Li\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$w = \frac{1}{2}Li^2 \qquad w = \frac{1}{2}Cv^2$$

- 1) Label terminals a and b
  - "Find the Thevenin or Norton equivalent circuit as seen by -----"
  - "Find the Thevenin or Norton equivalent circuit with respect to terminals a and b"
- 2) Find any components we can ignore (redraw if needed).
- 3) Decide which two of the three are we going to solve for: v_{oc}, i_{sc}, or R_{eq}
  - If there are dependent sources, solving for R_{eq} will require using a test source
- 4) v_{oc} circuit
  - Open across a and b, redraw, and solve for v_{oc}
- 5) i_{sc} circuit
  - Short across a and b, redraw, and solve for i_{sc}
- 6) R_{eq} circuit
  - Zero all your independent sources (short V-sources, open I-sources)
  - R_{eq} is the equivalent resistance as seen across a and b, or....
  - $R_{eq} = v_T/i_T$ where a test source is placed between a and b (active sign convention between v_T and i_T)
  - Redraw and solve for R_{eq}

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)]\, e^{-(t-T_0)/\tau}$$
$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)]\, e^{-(t-T_0)/\tau}$$
(Switch action at $t = T_0$)

$$\omega = 2\pi f = \frac{2\pi}{T}$$
$$\frac{\phi}{360°} = \frac{\Delta t}{T}$$
$$\sin(\theta) = \cos(\theta - 90°)$$
$$\cos(\theta) = \sin(90° - \theta)$$

1st Order

ODE: $\frac{dx}{dt} + ax = b$, General solution: $x(t) = x(0)e^{-at} + \frac{b}{a}(1 - e^{-at})$

ODE: $\frac{dx}{dt} + ax = 0$, General solution: $x(t) = x(0)e^{-at}$

Note that: $\frac{b}{a} = x(\infty)$, $a = \frac{1}{\tau}$

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2nd Order

Table 6-2: General solution for second-order circuits for $t \geq 0$.

$x(t)$ = unknown variable (voltage or current)

Differential equation: $x'' + ax' + bx = c$

Initial conditions: $x(0)$ and $x'(0)$

Final condition: $x(\infty) = \frac{c}{b}$

$$\alpha = \frac{a}{2} \qquad \omega_0 = \sqrt{b}$$

Overdamped Response $\alpha > \omega_0$

$$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)]\, u(t)$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2} \qquad A_2 = -\left[\frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2}\right]$$

Critically Damped $\alpha = \omega_0$

$$x(t) = [(B_1 + B_2 t)e^{-\alpha t} + x(\infty)]\, u(t)$$

$$B_1 = x(0) - x(\infty) \qquad B_2 = x'(0) + \alpha[x(0) - x(\infty)]$$

Underdamped $\alpha < \omega_0$

$$x(t) = [(D_1 \cos\omega_d t + D_2 \sin\omega_d t)e^{-\alpha t} + x(\infty)]\, u(t)$$

$$D_1 = x(0) - x(\infty) \qquad D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Auxiliary Relations

$$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\[2mm] \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

Complex Power Equations:

$$\vec{S} = \overrightarrow{V_{rms}} \overrightarrow{I_{rms}}^* = |\overrightarrow{I_{rms}}|^2 \vec{Z} = P + jQ$$

$$P = \frac{V_m I_m}{2} cos\theta$$

$$Q = \frac{V_m I_m}{2} sin\theta$$

$$pf = cos\theta$$

$$rf = sin\theta$$

$i_c = C\frac{dv_c}{dt}$

$\Rightarrow V_c' = \frac{1}{C} i_c$

$i_L' = \frac{1}{L} V_L$